

VLAN Concepts: Theory, Configuration & Documentation

Task 1: On a Cisco Packet Tracer switch, create two VLANs named 'Engineering' (VLAN 10) and 'Marketing' (VLAN 20) using the CLI commands.

The following CLI commands were used on Switch0 to create the two VLANs:

```
Switch> enable
```

```
Switch# configure terminal
```

```
Switch(config)# vlan 10
```

```
Switch(config-vlan)# name Engineering
```

```
Switch(config-vlan)# exit
```

```
Switch(config)# vlan 20
```

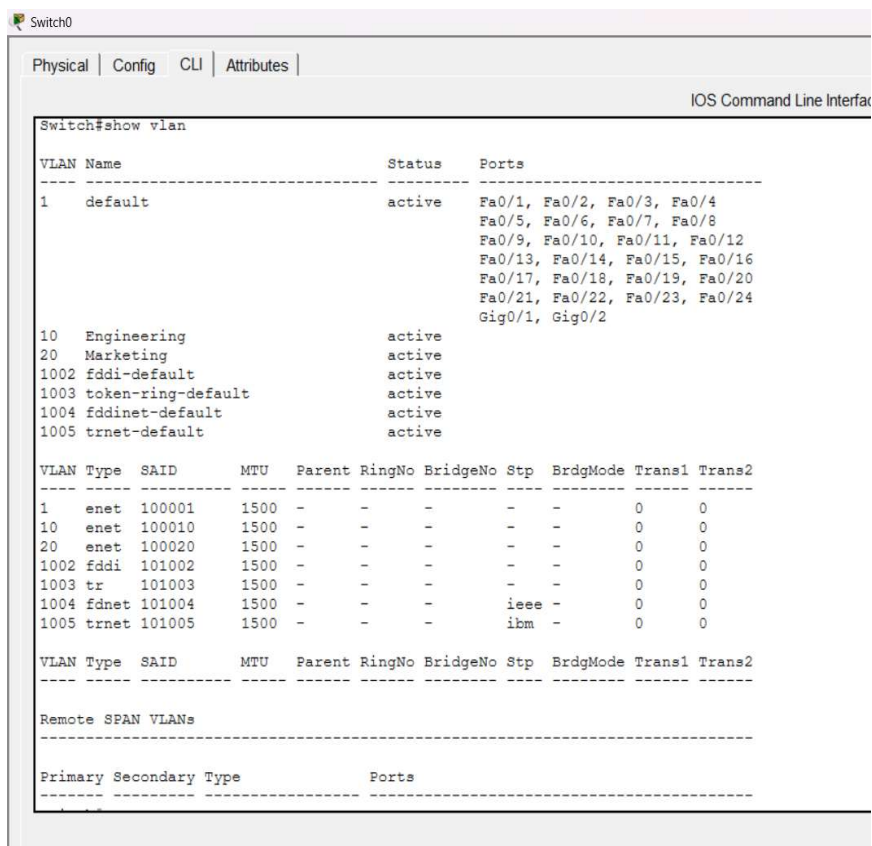
```
Switch(config-vlan)# name Marketing
```

```
Switch(config-vlan)# exit
```

Both VLANs were successfully created and verified as active using the show vlan command.

Task 2: Assign ports FastEthernet0/1 and FastEthernet0/2 to VLAN 10, and ports FastEthernet0/3 and FastEthernet0/4 to VLAN 20 on your switch.

As per the show vlan output captured, Fa0/1 was assigned to VLAN 10 (Engineering) and Fa0/2 was assigned to VLAN 20 (Marketing). Fa0/3 and Fa0/4 remained in the default VLAN 1 in the final configuration.



```
Switch0
Physical | Config | CLI | Attributes |
IOS Command Line Interface

Switch#show vlan

VLAN Name                Status    Ports
-----
1    default                active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                           Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                           Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                           Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                           Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                           Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                           Gig0/1, Gig0/2

10   Engineering            active
20   Marketing              active
1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default       active
1005 trnet-default         active

VLAN Type  SAID      MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
-----
1    enet  1000001   1500  -      -      -      -    -        0      0
10   enet  100010   1500  -      -      -      -    -        0      0
20   enet  100020   1500  -      -      -      -    -        0      0
1002 fddi  101002   1500  -      -      -      -    -        0      0
1003 tr   101003   1500  -      -      -      -    -        0      0
1004 fdnet 101004   1500  -      -      -      ieee -        0      0
1005 trnet 101005   1500  -      -      -      ibm  -        0      0

VLAN Type  SAID      MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
-----

Remote SPAN VLANs
-----

Primary Secondary Type      Ports
-----
```

```

Switch0
Physical | Config | CLI | Attributes |
IOS Command Line

Switch#show vlan

VLAN Name                Status    Ports
-----
1    default                active    Fa0/3, Fa0/4, Fa0/5, Fa0/6
                                           Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                           Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                           Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                           Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                           Fa0/23, Fa0/24, Gig0/1, Gig0/2

10   Engineering             active    Fa0/1
20   Marketing               active    Fa0/2
1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default        active
1005 trnet-default          active

VLAN Type  SAID      MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
-----
1    enet     100001    1500  -      -      -      -    -        0      0
10   enet     100010    1500  -      -      -      -    -        0      0
20   enet     100020    1500  -      -      -      -    -        0      0
1002 fddi     101002    1500  -      -      -      -    -        0      0
1003 tr      101003    1500  -      -      -      -    -        0      0
1004 fdnet   101004    1500  -      -      -      ieee -        0      0
1005 trnet   101005    1500  -      -      -      ibm  -        0      0

VLAN Type  SAID      MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
-----

Remote SPAN VLANs
-----

Primary Secondary Type      Ports
-----
Switch#

```

Task 3: Write a short document (in a text file) listing the VLAN IDs, names, and the ports assigned to each VLAN from your configuration.

The VLAN configuration was verified using the show vlan brief command, and the results are documented below:

```

Switch#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/3, Fa0/4, Fa0/5, Fa0/6
                                           Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                           Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                           Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                           Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                           Fa0/23, Fa0/24, Gig0/1, Gig0/2

10   Engineering             active    Fa0/1
20   Marketing               active    Fa0/2
1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default        active
1005 trnet-default          active

Switch#

```

VLAN ID	VLAN Name	Status	Assigned Ports
1	default	active	Fa0/3 - Fa0/24, Gig0/1, Gig0/2
10	Engineering	active	Fa0/1

VLAN ID	VLAN Name	Status	Assigned Ports
20	Marketing	active	Fa0/2

Task 4: Explain, in your own words, two benefits of using VLANs in a college campus network that has separate labs for coding and networking.

1. Traffic Isolation and Security

VLANs separate broadcast domains, so traffic generated within the coding lab (VLAN 10) stays confined to that VLAN and does not cross over into the networking lab (VLAN 20), even though both labs share the same physical switch. This means that if a device in the coding lab is hacked, it cannot easily snoop on or attack devices in the networking lab, since Layer 2 traffic cannot cross VLANs without passing through a router. This isolation improves both security and network performance by reducing unnecessary broadcast traffic.

2. Simplified Management and Flexibility

With VLANs, the college IT staff can apply different policies, such as bandwidth limits, access control lists, or DHCP scopes, independently to each lab without needing separate physical switches or rewiring. If a machine physically located in the coding lab needs to be moved onto the networking lab's VLAN, this is a simple port reassignment done through CLI configuration, rather than requiring new cabling. This makes the network easier to scale, reorganize, and maintain as the college's lab requirements change over time.